

FASER V10

Install, Build, Run — a consolidated reference

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- **ConvertGENIE / ConvertFASERMC**: convert a GENIE or official FASERMC neutrino-interaction sample into the common `TP0Event` format
- **FASERG4** (`faserps`): runs that through Geant4, writes per-event `TcalEvent` truth files
- **Batch** (`batchreco.exe`): reconstructs truth files into `TP0RecoEvent` files
- **EvDisplay** / analysis / **TauSearch**: consumes the reconstructed output
- `run_regression_tests.py`: wraps the whole chain end to end for an automated simulate → reconstruct → compare check

Known issue: FASERG4 scintillator proton fraction ($V9 \rightarrow V10$)

- FASERG4's scintillator material definition (3DCal tracker layers) had the wrong free-proton (H) vs. carbon-12 (C) content — fixed between production versions **V9** and **V10**
- **Root cause:** `G4Material::AddElement()` is overloaded — one variant takes an atom count (`G4int`), another a mass fraction (`G4double`). Declaring the stoichiometry counters as `G4double` silently selected the mass-fraction overload, so atom counts were read as mass fractions
- **Symptom:** free-proton and carbon-12 targets interacted at almost the same rate in V9 ($\approx 1:1$) instead of the physically expected $\approx 1:12$ ($\propto$ mass number A); V10 correctly gives $\approx 12:1$
- **Effect:** neutrino-induced CC/NC rates rise $\sim 15\text{--}18\%$ (a more isoscalar target supplies more neutrons for the neutrino-preferred d -quark coupling), antineutrino rates stay roughly flat; overall 3DCal fiducial total $+4.4\%$
- also shifts scintillator density ($1.03 \rightarrow 1.06 \text{ g/cm}^3$), hadronic interaction length, and dE/dx (Z/A change); Birks' constant has *not* yet been re-checked against the new composition
- full analysis: `FASERflux/faserg4_proton_fraction_bug.tex` (Fabio Cufino, André Rubbia)

Known issue: muon-spectrometer field consistency

- **Motivation:** the muon spectrometer's magnetic field was implemented *twice* — once in Geant4 simulation (`MuonMagneticField`) and once in GenFit reconstruction (`GenMagneticField`) — two independent hand-written implementations kept in sync only by hand, with no test proving they agreed and no check that either matched the real detector geometry
- **Fix:** extracted one shared, framework-independent function, `FASER::ComputeMuonSpectrometerField()`; both G4 and GenFit now call it as thin unit-converting adapters, so the two field *formulas* can no longer silently diverge — `MuonSpectrometerFieldTest (gtest)` checks G4 and GenFit agree
- **Real bug caught by the test:** outside the field's modelled envelope, GenFit's plain "Magnet" branch fell through to a stray non-zero fallback (0.001 kG in *Y*) where G4 returned an exact zero — a genuine, if tiny, pre-existing sim/reco divergence
- **Second problem:** the slit position deciding where the field flips sign / cuts to zero was a bare hardcoded literal duplicated in **four** places, agreeing only by convention, not construction — a future geometry change could silently desynchronize the field model from the geometry
- **Fix:** `MagnetGeometryProbe` samples the *actual* `TGeoShape` loaded from the real GDML to find where the iron truly ends and the slit truly begins, so reconstruction derives the slit position live instead of trusting a literal; `MagnetGeometryProbeTest` unit-tests the probe itself, including a case that deliberately mismatches the geometry to prove a future drift would actually be caught

Git workflow: sandboxed commit locks

- working from a sandboxed shell connected to the Mac, a `git commit` can leave stale `.git/index.lock` / `.git/HEAD.lock` / `.git/objects/maintenance.lock` files behind — the sandbox can create them but sometimes can't unlink them afterward (Operation not permitted)
- symptom: the *next* `git commit` fails outright with

```
fatal: cannot lock ref 'HEAD': Unable to create  
    '../.git/HEAD.lock': File exists.
```

- previously required manually running `rm -f` on the lock files from a real Terminal window on the Mac before the next commit could proceed
- **fixed:** a one-time delete-permission grant for the connected repo folder now lets stale locks be cleared directly from the same session — no manual Terminal step needed before any future commit

1. Install from scratch

```
git clone https://github.com/rubbiaa/FASER.git
cd FASER
source setup.sh
```

- `setup.sh` auto-detects the site (André's Mac, the Ubuntu ryzen01 box, or lxplus) and sources the matching `thisroot.sh` / `geant4.sh`
- hands off to `common_setup.sh`: CLHEP/Rave/GenFit paths, `LD_LIBRARY_PATH`, `CMAKE_PREFIX_PATH`, `PATH`, and a sanity check for missing/stale variables
- defines `fb` ("faser build"): reruns `cmake -build $HOMEFASER/build -j` from anywhere
- unrecognized machine? it prints exactly which `elif` branch to add
- full from-scratch ROOT/Geant4 install: `docs/INSTALL.md`, `docs/GEANT4_INSTALL.md`

2. Build

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=RelWithDebInfo
cmake --build build -j
```

- first build also compiles CLHEP, Rave, GenFit, googletest, and Pythia8 (CMake superbuild) — slow once, incremental afterward
- after source `setup.sh`, `fb` reruns the build step from anywhere
- executables land in `build/bin/`: `faserps`, `batchreco.exe`, `batchreco_detresp.exe`, `dumphits.exe`, `evDisplay.exe`, `ConvertGENIE.exe`, `Convert.exe`, `CombineFluxes.exe`, the `gtest` binaries, ...

CMake build options

Option	Default	Meaning
FASER_BUILD_EXTERNALS	ON	Build CLHEP/Rave/GenFit/ googletest/Pythia8 from source
FASER_BUILD_GEANT4_PACKAGES	ON	Build FASERG4 and FASERCalProtoG4 (needs Geant4)
FASER_BUILD_DISPLAY	ON	Build the GenFit-based event display / muon spectrometer package
FASER_BUILD_TAUSEARCH	ON	Build TauSearch
FASER_BUILD_TESTS	ON	Build FASER's own gtest suite, registered with CTest

Full list and reuse-an-existing-dependency examples: docs/INSTALL.md

3. \$FASERDATA

- every FASERG4/Batch executable resolves its data location through \$FASERDATA (FASER::GetDataDir(), in CoreUtils/FaserDataDir.hh) — not a hardcoded relative path or an inter-directory symlink
- GetDataDir() creates a requested subdirectory on demand if it doesn't already exist
- common_setup.sh defaults FASERDATA to \$HOMEFASER/data (gitignored) unless a per-site branch in setup.sh already exported it elsewhere (scratch, EOS, a dedicated checkout, ...)

\$FASERDATA subdirectories

Subdirectory	Contents	Written by / Read by
faserG4/	Per-event TcalEvent truth files	faserps / batchreco.exe, dumphits.exe, evDisplay.exe
batch/	Reconstructed TP0RecoEvent files	batchreco.exe / analysis code, summarize_output.py
GENIE/	GENIE input samples (too large for git)	fetch_data.py / faserps (--input-file)
GDML/	Detector geometry, exported every run	faserps / run_batchreco.py's default --geometry-file

4. `run_faserps.py` — runs the Geant4 simulation

- builds the V10-geometry macro in memory and pipes it straight to `faserps`' stdin — no `.mac` file to keep in sync
- full prose docs (custom geometry, interactive UI, MuonDIS physics): `run_faserps.md`
- default input sample is fetched automatically from a public CERNBox link the first time it's needed (sha256-verified)

run_faserps.py flags (1/2)

Flag	Default	Meaning
--build-dir	build/	CMake build dir containing bin/faserps
--vis	off	Launch the interactive Geant4 UI (no macro used)
--print-macro	off	Print the resolved macro text, exit
--dry-run	off	Print the command, don't execute
--input-file	default GENIE sample	ROOT input file; a relative custom value resolves against FASERG4/
--start-event	0	First event index to read
--n-events	100	Number of events (/run/beam0n)
--muons	off	Generate fixed-momentum muons instead of reading --input-file
--muon-momentum-gev	100.0	Muon momentum [GeV]

run_faserps.py flags (2/2)

Flag	Default	Meaning
--muondis	off	Enable on-the-fly Pythia8 DIS per muon interaction (implies --muons)
--muondis-cross-section-bias	150.0	Cross-section bias (1 = unbiased)
--muondis-q2min	1.0	Minimum Q^2 [GeV ²]
--muondis-interaction-log	disabled	CSV path logging each interaction
--muondis-pdf-set	built-in proton PDF	Path under FASER04/input/
--muondis-xbjmin	0.0	Minimum Bjorken-x cut
--muondis-debug	off	Verbose MuonDIS debug output
--tilt-deg	-4.5	Detector tilt about Y [deg]
--shift-x-cm / --shift-y-cm	45.0 / 24.0	LOS shift [cm]

run_batchreco.py — runs the reconstruction

- wraps Batch/batchreco.exe with named, validated flags instead of positional arguments
- full prose docs: run_batchreco.md

Flag	Default	Meaning
--run	<i>required</i>	Run number — deliberately no default
--min-event / --max-event	0 / 99999999	Event index range [min, max)
--mask	none (all events)	nueCC, numuCC, nutauCC, nuNC, or nuES
--multi-thread	off	Pass -mt to batchreco.exe
--geometry-file	\$FASERDATA/GDML/ FASERCAL_V10.gdml	GDML geometry to load
--build-dir	build/	CMake build dir containing bin/batchreco.exe
--split	1	Split the range into N parallel background jobs
--dry-run	off	Print the command(s), don't execute

run_regression_tests.py — simulate → reconstruct → compare

- runs the real pipeline end to end for several cases (nueCC/numuCC/nutauCC/nuNC, plus --muons and --muondis)
- compares aggregate output statistics against a committed golden JSON baseline

```
python3 run_regression_tests.py --record # first time / deliberate change  
python3 run_regression_tests.py # every other time: compare
```

run_regression_tests.py flags

Flag	Default	Meaning
--case	all cases	Restrict to a golden case name (repeatable); see --list
--record	off	Write the golden file(s) instead of comparing
--rel-tol	10^{-9}	Relative tolerance for floating-point comparisons
--build-dir	build/	CMake build directory
--python	current interpreter	Interpreter used for the subprocess calls (point at a PyROOT-enabled one if needed)
--list	—	List available case names, exit

fetch_data.py & summarize_output.py

`fetch_data.py` — public-CERNBox-link downloads into `$FASERDATA`, sha256-verified, used automatically by `run_faserps.py`.

Flag	Default	Meaning
<code>--force</code>	off	Re-download even if already present
<code>--dry-run</code>	off	Show what would be fetched, don't fetch
<code>--list</code>	—	List manifest entries, exit

`Tests/regression/summarize_output.py` — PyROOT summarizer used internally by `run_regression_tests.py` (requires `--truth-dir` and/or `--reco-file`).

Flag	Default	Meaning
<code>--dict-path</code>	CoreUtils' built dict	Path to the built ROOT dictionary
<code>--truth-dir</code>	none (skip)	Dir of truth files; needs <code>--run</code> , <code>--n-events</code>
<code>--reco-file</code>	none (skip)	Path to one Batch-TP0Recevent_*.root file

5. A typical end-to-end run

```
source setup.sh
cmake -S . -B build -DCMAKE_BUILD_TYPE=RelWithDebInfo && cmake --build build -j

python3 run_faserps.py --n-events 500 # simulate (auto-fetch)
python3 run_batchreco.py --run 10000 --mask numuCC # reconstruct numuCC
build/bin/evDisplay.exe 10000 numuCC # look at the result
```

or, to exercise the whole pipeline and check it against the committed baseline:

```
python3 run_regression_tests.py
```

6. Tests

```
ctest --test-dir build --output-on-failure
```

- FASER's own gtest suite (MuonSpectrometerFieldTest, MagnetGeometryProbeTest, ...) builds automatically whenever FASER_BUILD_TESTS and FASER_BUILD_GEANT4_PACKAGES are both ON (the default)
- the heavier simulate-reconstruct-compare regression suite is run_regression_tests.py, described above and in docs/REGRESSION_TESTS.md

- docs/INSTALL.md — the full CMake build: prerequisites, every option, troubleshooting a clean build
- docs/GEANT4_INSTALL.md — building/installing Geant4 itself
- run_faserps.md / run_batchreco.md — prose documentation for those two wrappers
- docs/README_MuonDIS.md — MuonDIS physics and every /physics/muondis/... option
- docs/REGRESSION_TESTS.md — the regression suite's full design, what's verified vs. not, and open items